

Typed contracts and schemas

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Architecture and data flow

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Typed contracts and schemas. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Typed contracts and schemas. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Classify counts 0, -1, "3", False under the stated contract.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Input `room_id=B`, `requested_count=true` passes a naive `isinstance(value, int)` check in Python. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Schema-valid input can still be unauthorised or semantically unsuitable; subsequent checks remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Classify counts 0, -1, "3", False under the stated contract.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled typed contracts and schemas evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of typed contracts and schemas. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Typed contracts and schemas

A labelled sample record. A lab tool accepts a string `sample_id` and an integer count of at least zero.

$A = \{\text{sample_id: "jar-8", count: 4}\}$; $B = \{\text{sample_id: "jar-9", count: "4"}\}$. No type conversion is allowed.

AI ACADEMY / AI-6/3

Typed contracts and schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect A

The fields and integer count satisfy the contract.



2. Inspect B

The quoted count is a string, so B fails the type check.



3. Return status

Accept A; reject B before running the lab operation.

CHANGE ONE THING

Change A count to -1.

Explain the changed result using the matching lesson.

Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept A; reject B before running the lab operation.

1. Inspect A
2. Inspect B
3. Return status

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Return status Accept A; reject B before running the lab operation. A schema checks declared structure. Looking like a number is different from having the required numeric type.

Practice 2: explain the deciding evidence

Changed-case practice: Change A count to -1. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Reject A on the non-negative value rule.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject A on the non-negative value rule. The type is correct but the value is outside its accepted range.

Repair a missing building block

Review Architecture and data flow (ai-6/2). Goal: Explain the mechanism and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
